Fiscal Year:	FY 2014	**Task Last Updated:**	FY 08/25/2015
PI Name:	Hughes-Fulford, Millie Ph.D.		
Project Title:	Antigen-Specific CD4+ T Cell Priming and Memory Response During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:	PI is deceased as of February 2021		
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	04/15/2009	**End Date:**	07/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131 ; ISS NOTE: Extended to 7/31/2014 per A. Chu/ARC and NSSC information (Ed., 4/3/2014) NOTE: New end date is 4/13/2014 per A. Chu/ARC and NSSC (Ed., 2/21/2013) NOTE: New end date is 4/13/2013 per A. Chu/ARC (Ed., 1/27/2012) NOTE: New end date is 4/14/2012 per A. Chu/ARC (Ed., 3/10/2011) NOTE: Grant number corrected to NNX09AH21G (from NNX09AG21G) per A. Chu/ARC (4/2010) NOTE: End date changed to 4/14/2011 (from 4/14/2012) per A. Chu/ARC (7/2009)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AH21G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Spaceflight immunosuppression is a significant obstacle to long-term human space travel. Of foremost concern is whether astronauts/cosmonauts may be able to generate effective protective immune responses against infections while in spaceflight. Using an innovative mouse experimental model, we propose a set of experiments that will test 1) whether initial specific activation of T cells is intact and 2) whether memory T cell function is maintained during spaceflight. Memory T cells are the mediators of immune protection after vaccinations and whether they function properly during spaceflight will determine if pre-flight vaccinations may be useful interventions. In our experimental system, transgenic T cells specific for a model protein will be transferred into recipient mice that have implanted mini-pumps. The mini-pumps will release the model protein along with non-infectious purified bacterial cell wall components. Specific T cells stimulated under these inflammatory conditions will be similar to those activated in a bacterial infection but no infectious agents will be involved. Mini-pumps will delay T cell stimulation until after mice acclimate to spaceflight to ensure that initiation of T cell activation occurs in microgravity. Naïve (previously unactivated) T cells will be transferred to determine whether responses to new immunological challenges (e.g., new infections) are intact. Previously activated T cells that will develop into memory cells will be transferred to determine whether robust secondary responses may be elicited during spaceflight. Our proposal directly addresses the research priorities of the Bion-M1 mission (changed to a Space Shuttle mission in 2010) to investigate resistance to infections and acquired immune responses without using infectious agents onboard the spacecraft. Together, these proposed experiments will address some of the most pressing outstanding questions in spaceflight immunology and more clearly define the risks and potential countermeasures of spaceflight immunosuppression.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The alteration of the immune system in spaceflight has been suggested by some to have many analogies to changes in the immune system during aging. In these studies of the immune system in mice, we have found that the changes in the immune function in flown mice are much the same as in humans. This suggests that there are common mechanisms of spaceflight immunosuppression in both mammals. Discoveries made from this study will help us understand the immune system and the basic initial steps of activation as well as help find new targets to use in the treatment of immunosuppression observed in aging and disease.
Task Progress:	Over 50% of the Apollo astronauts had bacterial or viral infections during flight or within 1 week of landing. Subsequent studies demonstrated that T-cells had a blunted activation response in vivo and in vitro after spaceflight. Studies from this lab have shown that early T-cell activation changes in microgravity. The objective of this study was to evaluate the effect of spaceflight on early T cell activation. C57BL/6J female mice were exposed to microgravity for 15 days on STS-131. Splenocytes were harvested within 2-3 hrs post-flight, cells isolated and activated (~9 hrs after landing) with Concanavalin A and antiCD28 for 2.5 hrs. Cells were harvested (11.5 hrs post landing) and stored in RNAprotect (Qiagen) at -80°C. Total RNA was isolated from the activated samples with RNeasy (Qiagen). We found that T-cell activation markers Interleukin-2 (IL-2), IL-2R alpha, interferon gamma (IFN gamma), TNF alpha, and IL-17 using quantitative real time RT-PCR. Expression of IL-2, IL-2R alpha, IFN gamma, CCL-3, TNF alpha, and IL-17 mRNA levels were significantly decreased 2-10 fold in the activated flight group as compared to ground controls. Expression of cyclophilin, and VEGF was unchanged. Protein synthesis of IL-2, IL-2R alpha, IFN gamma, CCL-3, TNF alpha, and IL-17 was decreased in the flight cell supernatants when compared to ground controls. The observed downregulation of induction of T-cell activation genes combined with changes of synthesis of these proteins show that animal immune cells require gravity and that the spaceflight induced changes in T-cell function remains after landing. More importantly, these data correspond with previous findings of human immune function in astronauts after return to Earth and support the conclusion that there is a physiological dependence of the immune system on gravity. This suggests that at least one mammalian system developed on this planet is dependent on gravity for full function. Immune function in mice splenocytes was significantly depressed, similar to the immunosuppressive changes seen in spaceflown T cells. Additionally, gene array data from spaceflown animals showed a similar pattern of immunosuppression as compared to animals on the ground with down regulation of several immune genes of interest. Expression of these genes was confirmed with qRTPCR. Activation was shown to be reduced in flown animals. Finally, the chemokines and cytokines associated with T cell activation were significantly increased with activation in ground animals and were suppressed in the flown animals. The lab is currently writing manuscripts on the data, so details of the data are not available until publication. Impaired induction of T-cell activation genes as well as the altered synthesis of immune chemokines and cytokines show that gravity is required for optimal immune function in mice. The spaceflight induced changes in T cell function remained for many hours after landing, suggesting that µg induced changes in the splenocytes require time to repair. These data suggest that at least one mammalian system developed on this planet is dependent on gravity for full function. More importantly, these data correspond well with our previous findings of inhibited human immune function in astronauts after return to Earth and support the conclusion that there is a physical dependence of the immune system on gravity. This project brings us one step closer to understanding the Earth's gravity influence on the immune system.
Bibliography Type:	Description: (Last Updated: 03/10/2018)
Articles in Peer-reviewed Journals	Chang TT, Spurlock SM, Candelario TL, Grenon SM, Hughes-Fulford M. "Spaceflight impairs antigen-specific tolerance induction in vivo and increases inflammatory cytokines." FASEB J. 2015 Jun 17. [Epub ahead of print] http://dx.doi.org/10.1096/fj.15-275073 ; PubMed PMID: 26085131 , Jun-2015
Articles in Peer-reviewed Journals	Martinez EM, Yoshida MC, Candelario TL, Hughes-Fulford M. "Spaceflight and simulated microgravity cause a significant reduction of key gene expression in early T-cell activation." Am J Physiol Regul Integr Comp Physiol. 2015 Mar 15;308(6):R480-8. Epub 2015 Jan 7. http://dx.doi.org/10.1152/ajpregu.00449.2014 ; PubMed PMID: 25568077; PubMed Central PMCID: PMC4360066 , Mar-2015

Articles in Peer-reviewed Journals	Hughes-Fulford M, Chang TT, Martinez EM, Li CF. "Spaceflight alters expression of microRNA during T-cell activation." FASEB J. 2015 Aug 14. [Epub ahead of print] http://dx.doi.org/10.1096/fj.15-277392 ; PubMed PMID: 26276131 , Aug-2015